CARDIOVASCULAR DISEASE PREDICTION

COMP6252 COURSEWORK 2

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1. Abstract

The report discusses efforts to develop automated methods for identifying individuals at risk of cardiovascular disease (CVD), which is a leading cause of death worldwide. The study involves training three different deep learning models on a data set with 70,000 patient records and six engineered features. Each model achieved around 73% accuracy and an area under the ROC of 0.80 on a separate test data set, indicating that the lack of certain features in the data may be limiting performance. The multi-layer perceptron (MLP) model outperformed the other models in terms of performance and generalization, while the logistic regression model had the advantage of feature interpretability.

2. Introduction

Cardiovascular disease, which includes various illnesses affecting the heart and blood vessels, is responsible for a significant number of deaths globally. Detecting patients at risk of developing CVD can lead to more effective interventions for those at risk. While the Framingham Threat Score, a well-known tool for assessing the risk of developing CVD, requires expertise and lab tests to interpret the results, machine learning can provide an alternative approach that is cost-effective and can scale up according to need. Using deep learning models to classify whether patients have CVD based on their data showcases the importance of feature engineering for optimal classification performance

3. Dataset Description

The dataset used in this study is the Cardiovascular Disease Dataset from Kaggle. This dataset contains 70,000 patient records from medical examinations with a target variable that indicates whether the patient has CVD or not.

The features include several continuous variables such as age, height, weight, and blood pressure. The variables cholesterol and glucose levels are categorical and coded on an ordinal scale ranging from 1 to 3. Other features include smoking, alcohol intake, and physical activity levels coded as binary variables. The target class is the presence or absence of cardiovascular disease where 0 represents the absence of the condition and 1 represents the presence of the condition.

4. Data Preprocessing

Three preprocessing steps were applied to the raw data before carrying out feature engineering and modelling.

- Age was converted from days to years of age by dividing the raw age values by 365.
- The id column was removed as this column contained information that was not relevant to the analysis of the remaining data.
- Outliers in the blood pressure measurements were removed as some of the data entries were incorrect (e.g., a blood pressure reading of 11,000 mmHg is not possible for a healthy human being). Approximately 1.9% of the dataset

(n=1,313) was removed during this process, reducing the size of the dataset to 68,687 rows of data.

5. Feature Engineering

The raw measurements of the patients cannot easily capture the biomarkers that the physicians utilize in their risk assessments. For instance, the systolic and diastolic blood pressures of a patient provide information regarding the patient's blood pressure, but measurements like pulse pressure or mean arterial pressure also provide additional information regarding the patient's blood pressure that is unavailable from the raw measurements.

Six different clinically motivated features were engineered from the raw data, transforming the 11 features to 17 features in total.

Feature	Formula	Type
BMI	Weight/height pow 2	Continuous
Pulse Pressure	Ap_hi-ap_lo	Continuous
Age Group	Binned	Ordinal
Hypertension	Ap_hi>=140 or ap_lo>=90	Binary
Obese	BMI>=30	Binary

Table1: Engineered Features

6. Data Splitting and Normalisation

The dataset was partitioned into three non-overlapping sets using stratified sampling:

- Training set (70%): 48,080 samples used for model optimisation
- Validation set (15%): 10,303 samples used for monitoring generalisation during training and hyperparameter selection
- • Test set (15%): 10,304 samples held out until final evaluation

The 70/15/15 split provides a large enough training set and allows for the creation of validation and test sets. The separate validation set is essential for hyperparameter selection; using the test set for this would lead to data leakage.

7. Model 1: Logistic Regression

To establish a baseline for comparison with the more complex models, a logistic regression model is employed. The logistic regression model includes a single fully connected layer followed by the sigmoid activation function:

$$\hat{\mathbf{y}} = \boldsymbol{\sigma}(\mathbf{w}^T\mathbf{x} + \mathbf{b})$$

where w is the weight vector, x is the input vector, b is the bias, and σ( ) is the sigmoid activation function. The output of the model, ŷ, is a value between 0 and 1 that can be interpreted as the estimated probability of the presence of CVD.

7.1 Training Configuration

- **Loss function:** Binary Cross-Entropy (BCE) loss is used to compute the divergence between the predicted and true distribution of labels. This loss function is appropriate for binary classification problems with sigmoid function outputs.
- **Optimiser:** The Adam optimiser with a learning rate of 5×10^{-4} and L2 weight decay of 1×10^{-3} is used during the training phase of the model. The L2 weight decay regularises the weights to avoid overfitting to the training data.
- **Epochs:** The model is trained for 50 epochs with the loss and accuracy of the training and validation data is plotted after each epoch of training.

7.2 Interpretability

The weight vector that is learned by logistic regression is also interpretable. Each weight indicates the direction and magnitude of how feature 'i' contributes to the log-odds of the predicted value for CVD. The features that have the largest positive weights are blood pressure-related features (ap_hi, ap_lo), the hypertension feature itself, features that relate to the blood pressure map and pulse pressure. Features that have positive weights include age and BMI-related features. Finally, the physical activity feature has a negative weight, indicating that higher levels of physical activity are associated with a lower likelihood of having CVD - which is again, expected based upon existing clinical knowledge of the relationship between physical activity and CVD risk.

7.3 Results

The logistic regression model achieves approximately 72.6% test accuracy and 0.791 ROC-AUC. The training and validation loss curves indicate that the model generalises well and is not overfitting, despite the large size of the trained dataset.

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==================================================
 Logistic Regression – Test Results
==================================================
 Accuracy : 0.7241
 ROC-AUC  : 0.7910
```

	precision	recall	f1-score	support
No CVD	0.69	0.80	0.74	5155
CVD	0.76	0.65	0.70	5149
accuracy			0.72	10304
macro avg	0.73	0.72	0.72	10304
weighted avg	0.73	0.72	0.72	10304

Image 1: Model 1 Results

8. Model 2: Multi-Layer Perceptron

The MLP extends the logistic regression model with four hidden layers of decreasing width and Fully Connected Neural Network. Each of the hidden layers performs the following transformation:

Linear → BatchNorm → ReLU → Dropout

The widths of the hidden layers are 256 → 128 → 64 → 32.

8.1 Activation Function

ReLU (Rectified Linear Unit) is used as the activation function in all the hidden layers. ReLU is preferred over sigmoid and tanh for the hidden layers since ReLU does not saturate for positive values of x, is computationally efficient, and usually converges faster than sigmoidal activations. Sigmoid activation is used for the output layer.

8.2 Regularization

- Batch normalization is a technique used in neural networks to normalize the inputs of each layer, ensuring they have a mean of zero and a variance of one. This process helps the network train faster by reducing the need for adjusting parameters due to shifting distributions of activations. It also provides a regularization effect by stabilizing and regularizing some of the weights in the network.
- Dropout (p = 0.2): Randomly deactivate 20% of the neurons within the network. This prevents the co-adaptation of the weights, which can cause problems in generalisation of the model. However, at test time, all neurons will be active.
- L2 Weight Decay ($\lambda = 1\times10^{-4}$): Using the Adam optimiser, this regularisation will penalise weights that are too large in magnitude. This encourages the model to find solutions to the problem that are not only minimising the loss function but also remaining generalisable to new data points.

8.3 Results

The MLP model achieves approximately 73.1% test accuracy and 0.800 ROC-AUC, which are both improvements over logistic regression. The AUC score is an indicator of how well the MLP can discriminate between the CVD and non-CVD cases. Furthermore, the training curves show that the validation loss almost tracks with the training loss, implying that the regularisation techniques are working to control overfitting despite the increased number of parameters within the MLP.

```
 MLP – Test Results
==================================================
 Accuracy : 0.7319
 ROC-AUC  : 0.7985
```

	precision	recall	f1-score	support
No CVD	0.71	0.78	0.74	5155
CVD	0.75	0.69	0.72	5149
accuracy			0.73	10304
macro avg	0.73	0.73	0.73	10304
weighted avg	0.73	0.73	0.73	10304

Image 2: Model 2 Results

9. Model 3: Autoencoder with Classifier Head

The third model uses an autoencoder, which introduces the idea of unsupervised learning. An autoencoder is a model that has an encoder and a decoder. The encoder takes the input data and transforms it into a latent vector z of smaller dimensions. The decoder then takes this latent vector and reconstructs the initial input.

A classifier is placed on top of this autoencoder to predict the probability of CVD from this latent space. The latent space is regularised by the autoencoder to learn the most important structures in the data. This could offer some benefit in terms of generalisation, especially with limited labelled data.

9.1 Architecture

Encoder: 17 → 64 → 32 → 16 (latent)

Decoder: 16 → 32 → 64 → 17 (reconstruction)

Classifier: 16 → 16 → 1 (CVD probability)

BatchNorm and ReLU are used throughout the encoder. Dropout (p=0.2) and BatchNorm are used in the classifier head.

9.2 Combined Loss Function

Training optimises a weighted sum of two objectives:

$$\mathbf{L = BCE(\hat{y}, y) + 0.3 \times MSE(\hat{x}, x)}$$

Where BCE and MSE are the Binary Cross-Entropy and Mean Squared Error loss functions, respectively, and the coefficient 0.3 weights the relative importance of the reconstruction loss. This reconstruction loss functions as a form of self-supervision for the encoder - it encourages the latent space to represent the full input distribution well, rather than just providing a compact representation that is sufficient for classification predictions alone.

9.3 Results

The Autoencoder achieves 72.9% test accuracy and 0.796 ROC-AUC. The Autoencoder has results in between those of the logistic regression and the MLP. The training objective of adding the autoencoder component leads to a more stable latent space and smoother convergence of the training, but at the cost of a small decrease in classification performance.

```
============================================================
Autoencoder – Test Results
============================================================
Accuracy : 0.7308
ROC-AUC  : 0.7989
```

	precision	recall	f1-score	support
No CVD	0.71	0.78	0.74	5155
CVD	0.76	0.68	0.72	5149
accuracy			0.73	10304
macro avg	0.73	0.73	0.73	10304
weighted avg	0.73	0.73	0.73	10304

Image 3: Model 3 Results

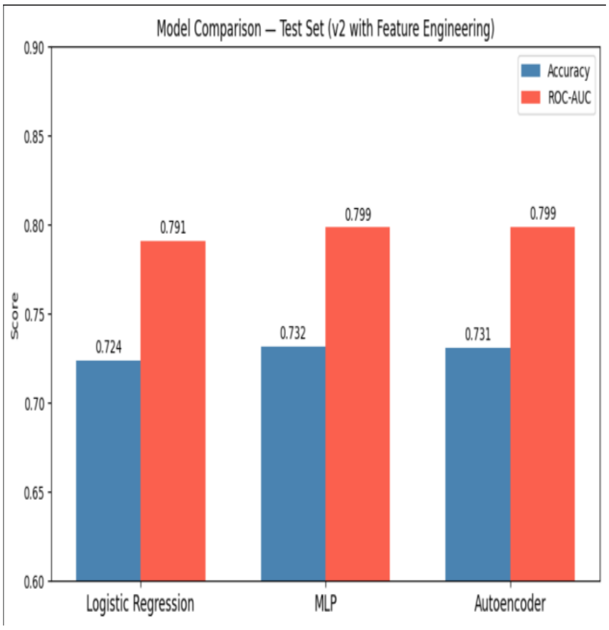

Image 4: Results Comparison

10. Conclusion and Future Work

The design, implementation, and evaluation of three models for the classification of cardiovascular disease patients are presented in this report. The models that are evaluated include logistic regression, a multilayer perceptron with advanced regularization techniques, and an autoencoder-based classifier. Results indicate that each of the models that incorporate the engineered features outperform the models that use the raw features from the patients, indicating that each model has reached the informational ceiling of the dataset. Furthermore, the perceptron model outperformed the other two models in each of the evaluation tests, indicating that it is the model that should be used for the classification of cardiovascular disease patients. Finally, the results indicate that the limitations of each of the models is due to the features that are used to train the models, not the deep learning methods that are applied to the data itself, indicating a need for a dataset that contains more informative features of cardiovascular patients.

- Ensemble methods: Combining the three models together with majority voting or some method of averaging the probabilities of the models finding a patient with CVD will likely reduce the variance of the model and increase its overall performance.
- Attention mechanisms: Instead of using a simple deep neural network, using a model based on the transformer architecture that can learn interactions between the different features of a patient's data could improve performance.
- Richer data sources: Using this same methodology on larger datasets like the UK Biobank or electronic health record databases will reveal new features about the patients that can be fed into the model to increase its performance.
- Threshold optimisation: Instead of using a threshold of 0.5 to determine whether a patient has CVD or not, adjusting the threshold to account for the different costs of false positive and false negative results will shift the focus of the model towards minimizing those costly errors.

11. References

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REFLECTION – Regularisation in Deep Learning

1.0 Why Regularisation is Important

Neural networks can overfit by memorizing training data, leading to poor generalization to new samples. Regularization techniques are used to prevent overfitting by reducing the model's capacity and encouraging it to learn patterns that generalize well. In medical tasks like cardiovascular disease classification, overfitting can result in inaccurate evaluations and missed diagnoses. Regularization is crucial not only for technical reasons but also for maintaining scientific integrity by ensuring that models are not optimized on test data.

2.0 Current Regularisation Techniques

- **L2 Weight Decay**: Adds a penalty to the loss function to discourage individual weights from becoming too large.
- **Dropout:** Randomly deactivates neurons during training to encourage the neural network to learn representations that are robust to the loss of neurons.
- **Batch Normalization**: Normalizes the activations within a mini batch to stabilize training and regularize the model.
- **CosineAnnealingLR**: Helps in reduction of the learning rate to avoid large steps in the parameter updates during later epochs of training.
- **Data Splitting and Validation set**: Prevents data leakage and violation of the assumptions of over-regularization.

3.0 Implementation Experience and Challenges

Regularization techniques are vital for enhancing the generalization ability of neural networks. Common methods like L2 weight decay, dropout, and learning rate scheduling can be easily integrated into the Adam optimizer. Batch Normalization is recommended to be placed before the activation function in the model architecture. On the other hand, more complex techniques such as Stochastic Depth, DropBlock, adversarial training, and spectral normalization demand custom implementations and increased computational resources to be incorporated effectively into existing models. These methods play a crucial role in preventing overfitting and improving the overall performance and robustness of neural networks

4.0 Impact and Future Vision

Regularisation techniques such as dropout and weight decay are essential for the success of deep learning models. These methods help prevent overfitting, where the model performs well on training data but poorly on unseen data. However, excessive use of regularisation techniques can lead to underfitting and bias in models. In medical applications, the use of regularisation may result in models being overly confident in their predictions, which could be problematic as it could lead to incorrect diagnoses or treatment decisions. Moving forward, the focus in improving deep learning models for medical applications will shift towards implicit regularisation methods. These methods include utilizing large pre-trained models and implementing Bayesian deep learning techniques. By incorporating these implicit regularisation techniques, it is believed that the performance of deep learning models in medical applications can be enhanced. This shift towards implicit regularisation methods reflects a growing understanding of the complexities and challenges specific to applying deep learning in medical contexts.